

Diagonalization of Matrices

1. Inverse of a Matrix

Definition

For an $n \times n$ matrix A , the **inverse** A^{-1} satisfies:

$$AB = BA = I_n$$

- A matrix with an inverse is **nonsingular (invertible)**.
- A matrix without an inverse is **singular (non-invertible)**.
- A^{-1} does **not** exist if and only if $|A| = 0$.

How to Compute A^{-1}

Set up the augmented matrix $[A \mid I_n]$ and apply row operations until the left side becomes I_n . The right side then becomes A^{-1} :

$$[A \mid I_n] \xrightarrow{\text{row ops}} [I_n \mid A^{-1}]$$

Shortcut for 2×2 matrices: If $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ and $ad - bc \neq 0$:

$$A^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

Key insight: Swap the main diagonal entries, negate the off-diagonal entries, divide by the determinant.

Examples

Example 1. $A = \begin{bmatrix} 1 & 2 \\ 4 & 5 \end{bmatrix}$

$|A| = (1)(5) - (2)(4) = -3 \neq 0$, so A^{-1} exists:

$$A^{-1} = \frac{1}{-3} \begin{bmatrix} 5 & -2 \\ -4 & 1 \end{bmatrix} = \begin{bmatrix} -5/3 & 2/3 \\ 4/3 & -1/3 \end{bmatrix}$$

Example 2. $B = \begin{bmatrix} 4 & -6 \\ -2 & 3 \end{bmatrix}$

$|B| = (4)(3) - (-6)(-2) = 12 - 12 = 0 \rightarrow$ **No inverse exists.**

2. Eigenvalues and Eigenvectors

Key Vocabulary

Term	Definition
Characteristic matrix	$A - xI_n$
Characteristic polynomial	$f(x) = \ A - xI_n\ $
Characteristic equation	$f(x) = 0$
Eigenvalue λ	A real root of the characteristic equation
Eigenvector $\mathbf{v}$	A nonzero vector satisfying $(A - \lambda I_n)\mathbf{v} = \mathbf{0}$
Eigenspace $S(\lambda)$	The set of all eigenvectors for a given λ

How to Find Eigenvalues and Eigenvectors

- Step 1.** Form $A - xI_n$.
- Step 2.** Compute $f(x) = |A - xI_n|$ and set it to zero.
- Step 3.** Solve $f(x) = 0$. The real roots are the eigenvalues.
- Step 4.** For each eigenvalue λ , solve the homogeneous system $(A - \lambda I_n)\mathbf{v} = \mathbf{0}$ to find the eigenvectors.

Example — 2×2 case

Let $A = \begin{bmatrix} 1 & 1 \\ -2 & 4 \end{bmatrix}$.

Characteristic polynomial:

$$f(x) = \begin{vmatrix} 1-x & 1 \\ -2 & 4-x \end{vmatrix} = (1-x)(4-x) - (1)(-2) = x^2 - 5x + 6$$

Characteristic equation: $x^2 - 5x + 6 = 0 \Rightarrow (x - 2)(x - 3) = 0$

Eigenvalues: $\lambda_1 = 2, \lambda_2 = 3$

Eigenvectors:

- For $\lambda_1 = 2$: Solve $(A - 2I)\mathbf{v} = \mathbf{0} \rightarrow \begin{bmatrix} -1 & 1 \\ -2 & 2 \end{bmatrix} \mathbf{v} = \mathbf{0} \rightarrow v_1 = v_2 \rightarrow \mathbf{v}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$
- For $\lambda_2 = 3$: Solve $(A - 3I)\mathbf{v} = \mathbf{0} \rightarrow \begin{bmatrix} -2 & 1 \\ -2 & 1 \end{bmatrix} \mathbf{v} = \mathbf{0} \rightarrow 2v_1 = v_2 \rightarrow \mathbf{v}_2 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$

Example — 3×3 with repeated eigenvalue

Let $C = \begin{bmatrix} 1 & -3 & 3 \\ 3 & -5 & 3 \\ 6 & -6 & 4 \end{bmatrix}$.

Characteristic polynomial: $f(x) = -(x - 4)(x + 2)^2$

Eigenvalues: $\lambda_1 = 4$ (simple), $\lambda_2 = -2$ (repeated, algebraic multiplicity 2)

The repeated eigenvalue $\lambda_2 = -2$ will be critical when checking diagonalizability (see below).

Multiplicity — Two Types

Type	Meaning
Algebraic multiplicity	The power of $(\lambda - x)$ as a factor in the characteristic polynomial
Geometric multiplicity	The dimension of the eigenspace $S(\lambda)$; equivalently, the number of linearly independent eigenvectors for λ

Geometric multiplicity $\leq$ Algebraic multiplicity (always).

3. Diagonalization of Matrices

Similarity of Matrices

Two $n \times n$ matrices A and B are **similar** if there exists an invertible matrix C such that:

$$B = C^{-1}AC$$

Similar matrices share the same eigenvalues, characteristic polynomial, and determinant.

What is Diagonalization?

An $n \times n$ matrix A is **diagonalizable** if it is similar to a diagonal matrix D :

$$D = P^{-1}AP$$

where D has the eigenvalues of A along its main diagonal, and P has the corresponding eigenvectors as its columns.

Steps to Diagonalize A

- Find all eigenvalues** $\lambda_1, \lambda_2, \dots, \lambda_n$ by solving $|A - xI_n| = 0$.
- Find eigenvectors** $\mathbf{v}_i$ for each λ_i by solving $(A - \lambda_i I_n)\mathbf{v} = \mathbf{0}$.
- Form matrix** P whose columns are the linearly independent eigenvectors: $P = [\mathbf{v}_1; \mathbf{v}_2; \dots; \mathbf{v}_n]$.
- Compute** P^{-1} .
- Verify** that $P^{-1}AP = D$, where $D = \text{diag}(\lambda_1, \lambda_2, \dots, \lambda_n)$.

When is A NOT Diagonalizable?

A matrix fails to be diagonalizable in two situations:

Condition 1. The characteristic equation has **non-real roots**.

Example: $E = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix}$ has eigenvalues $1, 1 + i, 1 - i$ — not all real → **not diagonalizable**.

Condition 2. Some eigenvalue has **geometric multiplicity < algebraic multiplicity**.

Example: $F = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$ has $\lambda = 0$ with algebraic multiplicity 2 but geometric multiplicity 1 → **not diagonalizable**.

Notable Fact

Every symmetric matrix is diagonalizable. (A matrix A is symmetric if $A = A^T$.)

Quick Summary Table

Condition	Diagonalizable?
All eigenvalues are distinct (real)	Yes
Repeated eigenvalue, geom. mult. = alg. mult.	Yes
Any non-real eigenvalue	No
Repeated eigenvalue, geom. mult. < alg. mult.	No
A is symmetric	Yes (always)

Worked Example: Full Diagonalization

Let $B = \begin{bmatrix} 0 & 0 & -2 \\ 0 & -2 & 0 \\ -2 & 0 & 3 \end{bmatrix}$.

Step 1 — Eigenvalues.

Characteristic polynomial: $f(x) = -(x + 2)(x - 4)(x + 1) = 0$

Eigenvalues: $\lambda_1 = -2, \lambda_2 = 4, \lambda_3 = -1$ (all distinct → diagonalizable)

Step 2 — Eigenvectors (solve $(B - \lambda I)\mathbf{v} = \mathbf{0}$ for each λ).

- $\lambda_1 = -2$: $\mathbf{v}_1 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$
- $\lambda_2 = 4$: $\mathbf{v}_2 = \begin{bmatrix} 1 \\ 0 \\ -2 \end{bmatrix}$

- $\lambda_3 = -1$: $\mathbf{v}_3 = \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix}$

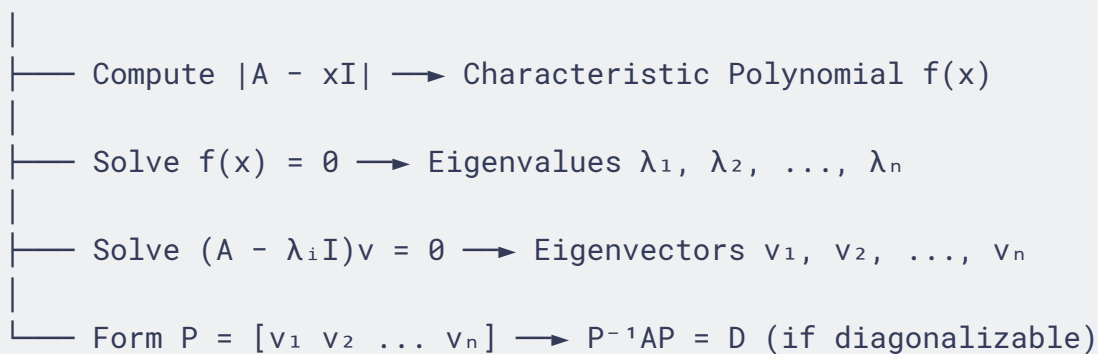
Step 3 — Form P and D .

$$P = \begin{bmatrix} 0 & 1 & 2 \\ 1 & 0 & 0 \\ 0 & -2 & 1 \end{bmatrix}, \quad D = \begin{bmatrix} -2 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & -1 \end{bmatrix}$$

Step 4 — Verify: $P^{-1}BP = D$ ✓

Concept Map

Matrix A



Reviewer compiled from: Reynera, M.D. — MATH 113 Linear Algebra, Chapter 5: Diagonalization of Matrices. Mariano Marcos State University.